

1) ABCB

2) ABBA

3) ABBA

$x = \underline{A} \underline{B} \underline{B} \underline{C} \underline{A} \underline{B}$

$y = \underline{A} \underline{B} \underline{C} \underline{B} \underline{A}$

$x \quad y$
 $P(\underline{A} \underline{B} \underline{B} \underline{C} \underline{A} \underline{B}, \underline{A} \underline{B} \underline{C} \underline{B} \underline{A})$

$\max(\overset{ABBA}{P(\underline{A} \underline{B} \underline{B} \underline{C} \underline{A}, \underline{A} \underline{B} \underline{C} \underline{B} \underline{A})}, \overset{ABCA}{P(\underline{A} \underline{B} \underline{B} \underline{C} \underline{A} \underline{B}, \underline{A} \underline{B} \underline{C} \underline{B} \underline{A})})$

$\overset{ABCB}{P(\underline{A} \underline{B} \underline{B} \underline{C} \underline{A} \underline{B}, \underline{A} \underline{B} \underline{C} \underline{B} \underline{A})}$

$\overset{ABB}{P(\underline{A} \underline{B} \underline{B} \underline{C}, \underline{A} \underline{B} \underline{C} \underline{B})} + \overset{ABC}{A}$

$\overset{ABC}{P(\underline{A} \underline{B} \underline{B} \underline{C} \underline{A}, \underline{A} \underline{B} \underline{C})} + B$

$\max(\overset{1}{P(\underline{A} \underline{B} \underline{B}, \underline{A} \underline{B} \underline{C} \underline{B})}, \overset{2}{P(\underline{A} \underline{B} \underline{B} \underline{C}, \underline{A} \underline{B} \underline{C})}) \max(\overset{3}{P(\underline{A} \underline{B} \underline{B} \underline{C}, \underline{A} \underline{B} \underline{C})}, \overset{4}{P(\underline{A} \underline{B} \underline{B} \underline{C} \underline{A}, \underline{A} \underline{B})})$

①

$\overset{ABB}{P(\underline{A} \underline{B} \underline{B}, \underline{A} \underline{B} \underline{C} \underline{B})}$

$\downarrow$
 $\overset{AB}{P(\underline{A} \underline{B}, \underline{A} \underline{B} \underline{C})} + B$

$\max(\overset{A}{P(\underline{A}, \underline{A} \underline{B} \underline{C})}, \overset{AB}{P(\underline{A} \underline{B}, \underline{A} \underline{B})})$

$\max(\overset{\square}{P(\underline{\square}, \underline{A} \underline{B} \underline{C})}, \overset{A}{P(\underline{A}, \underline{A} \underline{B})}) \quad \overset{A}{P(\underline{A}, \underline{A})} + B$
 $\downarrow$
 $\max(\overset{\square}{P(\underline{\square}, \underline{A} \underline{B})}, \overset{A}{P(\underline{A}, \underline{A})})$

②

$\overset{ABC}{P(\underline{A} \underline{B} \underline{B} \underline{C}, \underline{A} \underline{B} \underline{C})}$

$\downarrow$
 $\overset{AB}{P(\underline{A} \underline{B} \underline{B}, \underline{A} \underline{B})} + C$

$\downarrow$
 $\overset{A}{P(\underline{A} \underline{B}, \underline{A})} + B$

$\max(\overset{A}{P(\underline{A}, \underline{A})}, \overset{\square}{P(\underline{A} \underline{B}, \underline{\square})})$
 $\downarrow$
 A

③

$$\begin{aligned}
 & P(ABBC \overset{ABC}{ABC}), \\
 & \quad | \overset{AB}{AB} \\
 & P(ABB, AB) + C \\
 & \quad | \overset{A}{A} \\
 & P(AB, A) + B \\
 & \swarrow \quad \searrow \\
 & \max(\overset{A}{\cancel{P(A, A)}} \quad \overset{\square}{\cancel{P(AB, \square)}})
 \end{aligned}$$

④

$$\begin{aligned}
 & P(ABBCA \overset{AB}{AB}, AB) \\
 & \swarrow \quad \searrow \\
 & \max(P(ABBC \overset{AB}{AB}, AB) \quad P(ABBCA \overset{A}{A}, A)) \\
 & \swarrow \quad \searrow \quad \quad \quad | \overset{\square}{\square} \\
 & \max(P(ABB \overset{AB}{AB}, AB) \quad P(ABBC \overset{A}{A}, A) \quad P(ABBC, \square) + A) \\
 & \swarrow \quad \searrow \quad \quad \quad \swarrow \quad \searrow \\
 & \vdots \quad \quad \quad \vdots
 \end{aligned}$$